Calculation of Transformer Capacity Suitable for Induction Generator Operation

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Abstract—The proportion of using induction generators instead of synchronous generators in small-scale power plants where power generation output is not large is gradually increasing. Induction generators, like induction motors, are the biggest problem in that a voltage drop occurs due to a large inrush current when the grid is connected. Since the terminal voltage of an induction generator is usually low voltage, a voltage drop can greatly affect other equipment on the same branch line. In order to reduce the inrush current of the induction generator, a lot of research has been conducted through improvement of the starting method. Since inrush current is related to impedance, it is necessary to review the capacity of the transformer on the power side of the generator connected to the grid. In order to analyze the effect of voltage drop due to inrush current, the transformer capacity was calculated using the formula for calculating the capacity of the emergency generator capable of responding to the start of an induction motor. In this study, voltage drops at the primary and secondary sides of the transformer were calculated using a simplified equivalent circuit and an electromagnetic transient analysis program, respectively, for the transformer capacity obtained according to the inrush current ratio. As a result of the analysis, when connecting the induction generator to the grid, the magnitude of the inrush current did not differ in the transformer capacity. However, there is little difference in the voltage drop on the primary side of the transformer, but on the secondary side of the transformer, it was confirmed that the voltage drop was significantly different depending on the transformer capacity.

Keywords—Emergency Generator, Induction Generator, Small Hydro Power, Transformer, Voltage Drop

I. INTRODUCTION

As interest in energy security and environmental issues caused by global warming has recently increased, the development of clean energy resources using new and renewable energy has attracted much attention[1]. Small hydro power generation has a very stable output and higher efficiency than solar power generation or wind power generation among many renewable energy sources. Recently the number of cases where hydropower is reduced in size is gradually increasing in places such as water and sewage systems, moving beams, and reservoirs, instead of constructing large dams.

Synchronous generators and induction generators can be applied to power plants that produce electricity. When the output of the power plant is large, the synchronous power plant is mainly applied, but when the output is low, many induction generators which are advantageous in terms of price and maintenance are applied. In most cases, small hydropower plants are installed at the end or after the distribution line. The construction of a small hydroelectric power plant may be installed at the end or the latter stage of the distribution line, or may be installed on the premises of a power plant or a water and sewage facility. Induction generators generate large inrush currents when connecting the generated power to the grid. When an inrush current occurs, a large voltage drop occurs. The rated voltage of the induction generator used for small hydro power is mostly low voltage. This is because it connects to the same low voltage branch as other loads. So, a temporary voltage drop due to inrush current may stop the operation of other load equipment. When connecting the generated power to the grid, it goes through a transformer. The voltage on the secondary side of the transformer is low voltage, and the primary side is 22.9kV like on a distribution line.

The inrush current by the induction generator lasts very short, but if the voltage drop exceeds a certain range, the normal operation of the device may be difficult, so it is necessary to select a transformer capacity that can reduce the voltage drop.

In this study, we calculated the capacity of a transformer suitable for keeping the voltage drop within a certain range for other loads connected to the bus line, when the induction generator is directly started and connected to the grid. In order to select the capacity of the transformer suitable for maintaining the voltage drop due to the inrush current of the induction generator within a certain range, the analysis using the simple equivalent circuit and the transient analysis program were compared and analyzed with each other.

II. SYSTEM CONFIGURATION

Figure 1 shows the distribution line for supplying power from the substation to the consumer and the induction generator is connected to the end via a transformer.

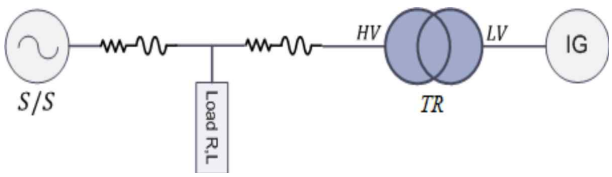


Fig. 1 System configuration diagram

Figure 2 is the result of investigating the capacity of a

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transformer used according to the output of an induction generator in a domestic small power plant. The capacity of the transformer is approximately 1.3 times the output of the induction generator. This is because the capacity of the transformer was determined regardless of the type of generator.

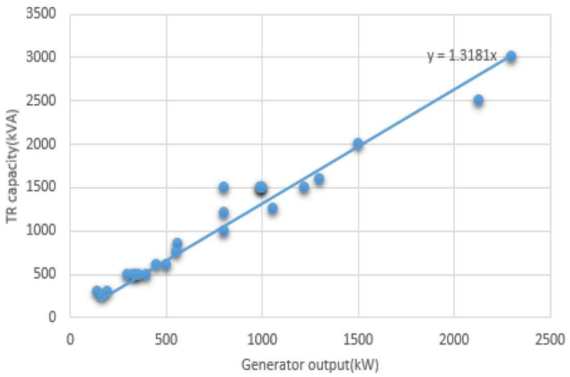


Fig. 2 Transformer capacity according to the output of the induction generator

Induction generators have a voltage drop component due to inrush current that does not appear in the synchronous generator when the grid is connected, so this should be considered when selecting the transformer capacity.

When these transformers are applied to an induction generator, it is necessary to analyze how the voltage drop due to inrush current varies. When replacing the circuit shown in Fig. 1 with an equivalent circuit in order to analyze the voltage fluctuations on the primary side and the secondary side of the transformer, it can be expressed as shown in Fig. 3.

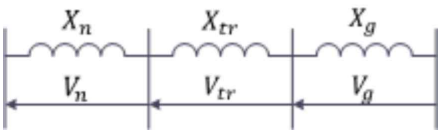


Fig. 3 Simplified equivalent circuit diagram for voltage drop analysis

Where X_n is the reactance of the distribution line, X_{tr} and X_g are the transformer and the induction generator, respectively, and V_n is the voltage at the distribution line, V_{tr} and V_g are the transformer and the induction generator, respectively.

In order to calculate the magnitude of the voltage drop generated when the induction generator is connected to the grid[9], each reactance must be obtained from the equivalent circuit in Figure 3. The reactance of a distribution line can be calculated as follows using the standard short-circuit capacity and terminal voltage of the power supply.

$$X_n = \frac{\left(\frac{V_t}{1000}\right)^2}{S_{sc}} \quad (1)$$

The reactance of a transformer can be obtained from its capacity and % impedance.

$$X_{tr} = \frac{\left(\frac{V_t}{1000}\right)^2 \times \%TR}{TR_{capacity}} \quad (2)$$

The reactance of the induction generator is obtained from the magnitude of the inrush current to the terminal voltage.

$$X_g = \frac{V_t}{I_{inrush}} \quad (3)$$

The magnitude of the voltage drop generated when the induction generator is connected to the grid is as follows.

$$V_{st} = \frac{X_g}{X_n + X_{tr} + X_g}[\%] \quad (4)$$

Therefore, the voltage drop rate caused by the inrush current is as follows[9].

$$V_{drop} = 1 - V_{st}[\%] \quad (5)$$

III. EMERGENCY GENERATOR AND TRANSFORMER

Emergency generators are installed to continuously provide stable power even in the event of a power outage. The capacity of the emergency generator is determined to be the largest among the capacity required for normal full-load operation, the capacity to cope with the starting of the motor of the maximum capacity, and the capacity to cope with the operation of the instantaneous maximum load. The magnitude of the inrush current of the induction generator is very similar to the starting current of the induction motor[]. The emergency generator capacity of the size that can respond to the voltage drop when starting the induction motor can be calculated as follows.

$$P_g = \left(\frac{1-\triangle E}{\triangle E}\right) \times x_d' \times Q_L \quad (6)$$

Where $\triangle E$ is the voltage drop rate, x_d' is the generator transient reactance, and Q_L is the inrush capacity at the start of the induction motor.

If the magnitude of the starting current of the induction motor and the inrush current of the induction generator are the same, the formula for calculating the capacity of the emergency generator by voltage drop is applied to the calculation of the capacity of the transformer connected to the induction generator as follows.

$$T_g = \left(\frac{1-\triangle E}{\triangle E}\right) \times x_d' \times k \times P_g \quad (7)$$

Where $\triangle E$ is the voltage drop rate, x_d' is the generator transient reactance, k is the inrush current to the rated current and P_g is the output capacity of the induction generator. Under the condition that the reactance of the induction generator and the ratio of the inrush current to the

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rated current are the same, the higher the ratio of the voltage drop rate, the lower the capacity of the transformer.

IV. ANALYSIS OF MEASUREMENT AND OPERATION CHARACTERISTICS

Table 1 shows the specifications of the induction generator applied for analysis in this study. The induction generator was selected as a deep-bar rotor that helps to reduce the magnitude of the current during inrush. The output of the induction generator that is installed and in operation uses low voltage 380V up to 200kW.

Table 1. Data of Induction Generator applied to simulation

Rating	200kW
Voltage	380V
No. of poles	4
Power factor	0.758
Efficiency	0.93

Assuming that the magnitude of the inrush current is 3-6 times the rated current, Table 2 shows the capacity of the transformer using equation (7). The reactance value of an induction generator depends on the voltage. In this study, since the voltage of the generator is less than 600V, the reactance value is 0.25[11]. The smallest value for finding the capacity of a transformer by adjusting the ratio of the inrush current of a 200㎾ induction generator from 3 to 6 times is 450㎸A, and the maximum is 900㎸A. The transformer capacity obtained as the inrush current is three times the rated current becomes 2.25 times the generator output. It can be seen that this is almost twice as high as the ratio of the transformer capacity to the output of the generator shown in Figure 2.

Table 2. Data of Induction Generator applied to simulation

Division	Multiple of Inrush Current			
	3	4	5	6
T_g	450 kVA	600 kVA	750 kVA	900 kVA

When a 3-phase 380V, 4-pole, 200㎾ induction generator is started at a distance of 10㎞ from the substation and connected to the system, the inrush current and voltage drop according to the capacity of the transformer were analyzed. A 5,000kW load with a power factor of 0.8 is connected to the distribution line connected to the induction generator in the substation.

Electromagentic transient analysis program[EMTP] was used to analyze how the voltage drop changes when transformer capacities differ for the same size inrush current Figure 4 calculates the magnitude of the inrush current, which depends on the magnitude of the transformer capacity(450㎸A;○, 600㎸A;□, 750㎸A;△, 900㎸A;×) when the induction generator presented in Table 1 is connected to the grid. It can be seen that the inrush current generated during grid connection is displayed about 5 times larger than the rating regardless of the change in the capacity of the transformer.

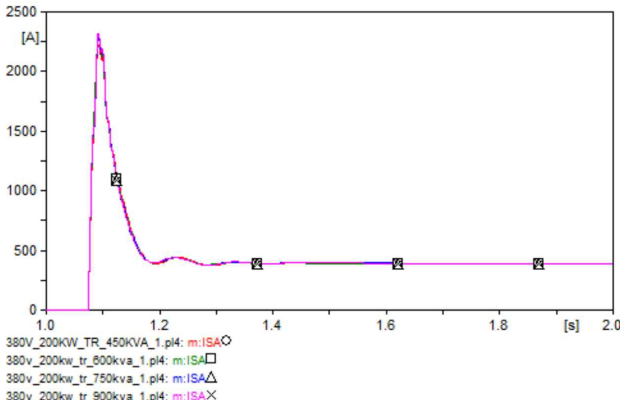


Fig. 4 Inrush current

Figure 5 shows the results of simulation how much voltage drop occurs on the primary side of the transformer when adjusting the magnitude of the capacity of the transformer(450㎸A;○, 600㎸A;□, 750㎸A;△, 900㎸A;×) when the induction generator presented in Table 1 is linked to the grid. The voltage drop rate that occurs at the extra-high voltage side when the induction generator is connected to the grid by direct on line starting is very low, with a difference of 0.8% even when adjusting the transformer capacity. This is because the voltage drop on the extra-high voltage side is very low even when an inrush current occurs, so the effect on other facilities is judged to be insignificant.

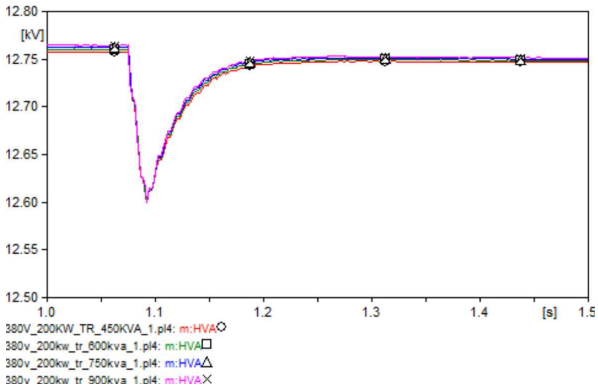


Fig. 5 Voltage variation on the extra high voltage side

Figure 5 shows the results of analysis of changes in the voltage drop that occur on the low voltage side when the magnitude of the transformer capacity(450㎸A;○, 600㎸A;□ 750㎸A;△, 900㎸A;×) is different when the induction generators presented in Table 1 are linked to the grid. It can be seen that when the capacity of the transformer is small, the voltage drop is displayed high, and as the capacity of the transformer is large, the voltage drop is displayed low.

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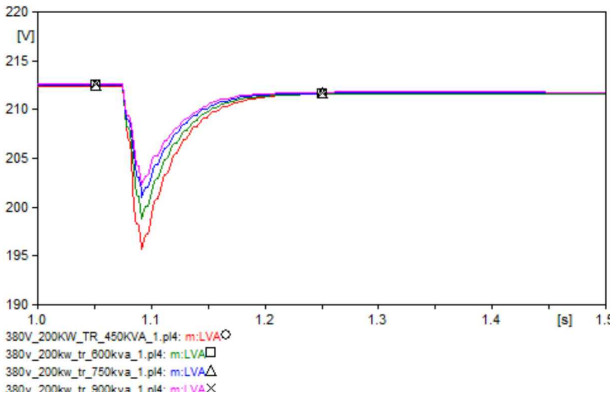


Fig. 6 Voltage variation on the low voltage side

As shown in Figure 6, when connecting a load such as an induction motor to the secondary branch line of a transformer, it is necessary to consider the effect of voltage drop. To reduce the effect of voltage drop, it is better to connect the load to the extra-high voltage side rather than the low-voltage side, improve the starting method of the induction motor, or select the transformer capacity to be more than a certain size than the output of the induction generator. The voltage drop due to the inrush current on the primary side of the transformer is almost negligible, but it appears very high on the secondary side, so the change in the voltage drop rate at the secondary side was calculated according to the size of the transformer capacity.

Table 3 shows the result of comparing the voltage drop rate at the secondary side obtained by using a simple equivalent circuit and an electromagnetic transient analysis program when the capacity of the transformer is adjusted by the same inrush current when connecting the induction generator to the system. As a result of analysis, it can be seen that the two calculation results are very close.

Table 3. Comparison of analysis result

TR capacity (㎸A)	Voltage drop(%)	
	Simple Equivalent Circuit Analysis	EMTP Analysis
450	7.8	7.8
600	6.6	6.5
750	5.4	5.4
900	4.7	4.8
1,200	3.7	3.8

V. CONCLUSION

In this paper, we studied to determine how much voltage drop occurs on other loads on the same bus line by inrush current when connecting the induction generator to the system. In order to calculate the capacity of the transformer suitable for reducing the voltage drop due to the inrush current, the calculation of the capacity of the emergency generator obtained in consideration of the direct start of the induction motor and the voltage drop was applied to this analysis. In order to analyze the degree of voltage drop due to inrush current, a simple analysis using an equivalent circuit and a transient analysis program were compared with each other. As a result of the analysis, it was found that the voltage drop rate was very close.

For the same inrush current, it was confirmed that the voltage drop on the primary side of the transformer hardly changed even with the change in the capacity of the transformer, but on the secondary side, the voltage drop significantly decreased as the transformer capacity increased

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